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RAG (Retrieval-Augmented Generation)

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Purpose

The purpose of RAG is to take the given prompt. Go out and retrieve useful information, depending on what you are asking it could use outside sources such as company documents or research papers. Augment which allows it to the AI's knowledge, making it more advanced. Then it generates and gives back a better response.

Scope

For this project, I will use outside resources such as research papers and case studies to show how RAG improves accuracy and reduces error in a large language model. This research will look at a deeper level of the way RAG works, what tools it takes to build one, and what effects it has, to fully understand how it works and why it is important.

Definitions & Abbreviations

Term/Acronym	Definition
RAG/ Retrieval-Augmented Generation	An AI method used to retrieve documents before answering. It doesn't need to look at its memory since it searches for the document first.
LLMs/ Large Language Models	An advanced AI system that is trained in text to understand and generate language. ChatGPT is an example.
Embedding	How a computer turns text into numbers so it can compare meetings. Use the numbers to see if they are similar.
Vector Database	A storage place for embedded numbers. Assists the AI to quickly find the most relevant pieces of information.
Hallucination	When AI gives an answer, that sounds right, but it is wrong.

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Research

Introduction

RAG is a way to improve how AI systems answer the given prompts. Large language models answer based on patterns that they have been programmed with. LLMs do not actually know the latest information, and there is no way of checking their work. This opens doors to problems since the model gives an answer that sounds right but is not supported by any true sources.

RAG changes those wrong answers by using AI to search through a collection of documents while it works. RAG lets the AI read first, answer after which reduces mistakes and the answers true and trustworthy. It also lets whoever is using it to control what information the AI can access.

How RAG Works

RAG uses 3 major stages. The first stage turns text into numbers using embedding. The documents need to be converted into a new way the computer can understand. RAG turns each document and the data in it into an embedding. An embedding is a long list of numbers that represent the meaning of the text. Since documents with similar meanings have embedded numbers that look similar. It will allow the AI to compare ideas, not just similar words.

The next step helps the user find the best matching documents. When the prompt is given, the system turns the question into an embedding. It will then search the database to find documents and find the embedding, which is most like the question asked. The process happens very quickly and pulls the most relevant information.

The final step is using the retrieved information to give the answer back to the user. It uses the document as a guide in order to produce the answer. The AI reads the text that it was given, uses it to connect further into the question, and then

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builds a response that has true information. Most RAG systems have the ability to use citations to help users see where it pulled the information from.

With all this combined, it makes a way for users to get reliable answers to all their questions.

Importance of RAG

The reason RAG is important is because it fixes one of the biggest issues with AI since the AI cannot check its own work. Since AI without RAG is only able to access and use what it was thought during training, which has a good chance of being outdated or missing information. Which allows the AI to produce answers that sound correct but are not. When an AI has a RAG, it searches the documents first which makes a big difference. It will help AI provide the correct answers based on the real information. RAG makes updates so much easier since you only will need to update the document database rather than training a new model every time new information comes out.

RAG also allows AI to be transparent and show the user where the AI pulled the information from. This allows the user to trust the AI's answers and verify them. Which makes RAG one of the most reliable and safer AI tools used today.

How RAG Supports Intel Work

RAG is very useful for intelligence since analysts are given huge amounts of information and finding the right details can take a while. RAG helps by searching all of the documents and giving back the user the most relevant information. For Open-Source Intelligence (OSINT), RAG is able to sort through all different kinds of sources to find trends and key details. When an analyst is writing a report, RAG is able to give correct sources which will save the analysts time.

RAG is also able to remember very well since analysts must understand past patterns to predict future events. Instead of looking through pages upon pages of

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research, RAG can be asked specific questions and answered properly. RAG helps analysts save time.

RAG Limitations

RAG is a very impressive and helpful tool which can allow users to save a lot of time; however, it does come with some limitations. One of the challenges that it has is that if you put bad documents into the AI, it will not run very efficiently. If the bad document contains mistakes or wrong facts, then the AI will give out wrong information. This means that the database that is given to the RAG is extremely important.

Another challenge happens when the RAG has to break down big documents into smaller chucks so the model can read them. This allows room for error since some text has been known to get lost when it is split up. Which causes the AI to miss parts of the correct answer and key details.

Security is also a big concern for RAG since if classified information is given to the database, it has to be protected. Without proper safeguards, the system could get classified information stolen and exposed.

Finally, the limitation with RAG is that it is still an AI tool. It is there to help you find and give organized information. Humans must still be liable to do the work and cannot rely on RAG and AI to do it for you.

The Future of RAG

RAG will be a very big use in the future. There are already many companies that are working on a safer and more trustworthy version with new features like automatic updates to keep the AI as smart as it can be. RAG looks like it will become the standard for AI, making people work smarter and faster. RAG is only a tool though there still needs to be human thinking which cannot be replaced.

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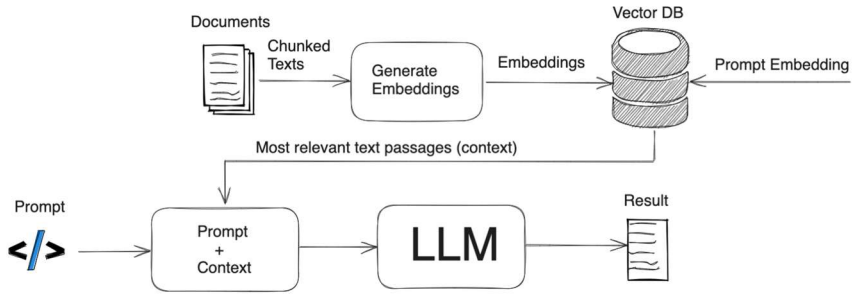
Sources

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